

Victor Trappler

Assistant Professor in Applied Mathematics

 [vtrappler.github.io](https://github.com/vtrappler)
 [vtrappler](#)
 [victortrappler](#)

Research interests

Assistant Professor at École des Mines de Saint-Étienne, my research interests revolve around **Multiobjective Optimisation under Uncertainties**, with an emphasis on **Bayesian Optimization**, **Uncertainty Quantification**, **Data Assimilation** and **Machine Learning**.

Education

2017–2021 **PhD in Applied Mathematics**, AIRSEA, Inria/LJK, Grenoble, France

Title: Parameter control in the presence of uncertainties

Abstract: Classical methods of parameter estimation usually imply the minimisation of an objective function, that overlooks the role of uncertain parameters. Strategies taking into account these uncertainties need to be defined.

Keywords: Parameter Estimation; Optimisation under Uncertainties; Gaussian Processes

Advisors: A. Vidard, É. Arnaud, L. Debreu

2015–2017 **MSc Mathematical Modelling and Computation**, Danmarks Tekniske Universitet, Kgs. Lyngby, Denmark

Double Degree with École Centrale Lyon

Focus points: Applied mathematical analysis, Dynamical Systems, Scientific Computing, Statistical modelling, Stochastic simulations

2013–2017 **Engineering Degree**, École Centrale Lyon, Écully, France

Interests and courses oriented toward applied mathematics

Double Degree with Danmarks Tekniske Universitet

Experience

Permanent position

2025– **Assistant Professor**, Institut Henri Fayol, Mines Saint-Étienne, Saint-Étienne, France

Assistant professor in Optimization, Uncertainty Quantification, Statistical Learning

Post-doctoral Positions

2024–2025 **Post-doctoral researcher**, Camille Jordan Institute (ICJ), Lyon, France

Title: Multiobjective optimisation under uncertainties

Abstract: Many real world decision problems can be summarized as multi objective optimization problems, where the sought quantity is the Pareto front. In the presence of uncertainties, many definitions of robustness can be defined, and appropriate numerical methods must be derived.

Keywords: Multiobjective optimisation under uncertainties, Uncertainty Quantification, Gaussian Processes

2021–2023 **Post-doctoral researcher**, Joint Laboratory Eviden/Inria, Grenoble, France

Title: Data Assimilation in latent spaces

Abstract: In this work, we propose to use ML in order to build state-dependent preconditioners for the inner loop in Variational Data Assimilation, and apply this method to a toy-model of a Shallow Water assimilation system. This work has led me to work on the different aspects and interactions of Data Assimilation, Machine Learning and Linear Algebra, within the R&D team "AI4Sim" of Eviden

Keywords: Data Assimilation; Uncertainty Quantification; Machine Learning; Linear Algebra

Internships/Master thesis

2017 **Master Thesis**, AIRSEA, Inria/LJK, Grenoble, France

Title: Parameter control in the presence of uncertainties: Robust estimation of bottom friction

Advisors: Uffe Høgsbro Thygesen (DTU), Élise Arnaud, Arthur Vidard, Laurent Debreu (Inria)

2015 **Intern, EDF R&D, Chatou, France**
Development of MATLAB tools for hydrodynamical model TELEMAC3D, application to residence time

Teaching and Supervising Experience

Supervisory Experience

2022 - **Exaucé Luweh Adjim Ngarti, Joint laboratory Eviden/Inria, PhD candidate**
Title: Deep Learning for Inverse Problems, application to oceanography
Robust Parameter Estimation Using Variational Inference and Generative Neural Networks

Teaching

2025 **Lecturer, Mines Saint-Étienne and ISTP**
○ Data Science
○ Statistics
○ Operations research and Decision support

2024 **Teaching assistant (24h), École Centrale Lyon**
Lectures in calculus for undergraduates.
○ L3: 24h of Applied Calculus for 1st year Engineering students

2017–2019 **Teaching assistant (138h), Grenoble-Alpes University**
Lectures in calculus, algebra, and computer lab sessions in statistics for undergraduates students:
○ L2 STA301: 90h of lab work on statistics using the R language for bio students.
○ L1 MIAHS: 20h of exercise sessions on calculus for social sciences students
○ L1 MAT104: 28h of lectures and exercise sessions on geometry and algebra for physics students

2017–2020 **Research and Teaching Label, Grenoble-Alpes University**
PhD completed with RES label: Specific doctoral training for students wanting to pursue an academic career, mostly on teaching methods and reflexions on higher education

Publications and Preprints

- 2024 Trappler, Victor and Arthur Vidard (2024). “State-dependent preconditioning for the inner-loop in Variational Data Assimilation using Machine Learning”. Submitted. URL: <https://hal.science/hal-04707967>.
- 2021 Trappler, Victor (June 2021). “Contrôle de Paramètre En Présence d’incertitudes”. PhD thesis. Université Grenoble Alpes. URL: <https://theses.hal.science/tel-03275015>.
- 2021 Trappler, Victor, Élise Arnaud, Arthur Vidard, and Laurent Debreu (2021). “Robust calibration of numerical models based on relative regret”. In: *Journal of Computational Physics* 426, p. 109952. ISSN: 0021-9991. DOI: <https://doi.org/10.1016/j.jcp.2020.109952>. URL: <https://www.sciencedirect.com/science/article/pii/S0021999120307269>.

Oral and Poster presentations

- 2024 **Talk:** Single and Multi-objective Bayesian Optimization under uncertainties – *Mexico Network meeting*
- 2024 **Talk:** Multiobjective Bayesian Optimization under uncertainties – *CIROQUO Days, CEA DAM*
- 2023 **Poster:** State-dependent preconditioning for VarDA – *9th International Symposium Data Assimilation, Bologna, Italia*
- 2023 **Poster:** State-dependent preconditioning for VarDA – *54th International Colloquium for Oceanography, Liège, Belgium*
- 2022 **Poster:** Regret-based estimates using GP – *CIROQUO scientific days, Grenoble, France*
- 2019 **Talk:** *Seminar of the Uncertainty Quantification Group, MIT, USA*
- 2019 **Talk:** *Applied Inverse Problems Conference, mini-symposium “Dimension reduction in inverse problems”, Grenoble, France*
- 2018 **Talk:** *National Colloquium for Data Assimilation, Rennes, France*
- 2018 **Poster:** *Workshop on Sensitivity Analysis and Data Assimilation in Meteorology and Oceanography, Aveiro, Portugal*

Languages and programming skills

Languages	French	<i>Fluent - Native</i>
	English	<i>Fluent - TOEFL IBT score: 105/120 (2015) and abroad experiences</i>
	German	<i>Adapted for casual conversations</i>
Sci. Comp.	Python	<i>Numpy, Scipy, Machine Learning (PyTorch, scikit-learn, MLFlow, Botorch)</i>
	R, Matlab, Julia, FORTRAN, C++	<i>Notions</i>
Utilities	$\LaTeX$, bash, git, Docker, slurm	

Miscellaneous

- 2024 **ORENI project**
Participation in some developments for the ORENI project (Optimisation of strategies of building refurbishment/construction under uncertainties <https://oreni.plateforme-tipee.com>)
- 2023 **UQ Working-Group, AI4Sim (Eviden)**
Launched and animated a working-group on Uncertainty Quantification applied to Geophysics within the AI4Sim team.
- 2022- **Individual Thesis Follow-up Committee**
Member of the Individual thesis follow-up committee (Comité Suivi Individuel) of Katarina Radišić (INRAE)
- 2020 **Representative of non-permanent employees, LJK, Grenoble**
Elected as a representative of the non-permanent employees (PhD, interns, postdocs fellows, engineers) of the Jean Kuntzmann Laboratory. Participation at the lab council